

# MUHAMMED IRFAN K

## MERN Stack Developer | Full Stack Developer

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### PROFILE SUMMARY

MERN Stack Developer with hands-on experience architecting scalable, real-time web applications using MongoDB, Express.js, React.js, and Node.js. Proven ability to design secure REST APIs, role-based authentication (JWT), payment integrations (Stripe), and real-time features (Socket.IO). Skilled in clean architecture, state management (Redux Toolkit, Context API), and performance optimization, with a track record of independently shipping and deploying production applications end-to-end.

### TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), TypeScript, Java, C, HTML5, CSS3
- **Frontend:** React.js, Redux Toolkit, Context API, React Router, Tailwind CSS, Bootstrap, GSAP (animations)
- **Backend:** Node.js, Express.js, RESTful API design, MVC & Clean Architecture
- **Database:** MongoDB (Mongoose, TTL indexes, Aggregation), MySQL
- **Integrations:** JWT Authentication, Socket.IO, Stripe Payments & Webhooks, Cloudinary, Nodemailer/Brevo SMTP, react-leaflet
- **Tools & Deployment:** Git, GitHub, Postman, Vercel, Render, npm
- **Practices:** Data Structures & Algorithms (LeetCode), Agile workflow, Code Review, Debugging & Testing

### EXPERIENCE

#### MERN Stack Developer Intern — Luminar Technolab

Aug 2024 – Mar 2025

Kochi, Kerala

- Engineered fully responsive MERN stack applications using MongoDB, Express.js, React.js, and Node.js, prioritizing performance optimization and scalable architecture.
- Architected and consumed RESTful APIs supporting real-time data flow and complete CRUD functionality across multiple modules.
- Streamlined team collaboration and version control using Git/GitHub, and validated backend endpoints through rigorous Postman testing and debugging.

### PROFESSIONAL DEVELOPMENT

#### Independent Project Building & Skill Development

Apr 2025 – Present

- Independently architected and shipped VoltSpot end-to-end, expanding it from a booking MVP into a production system with Stripe webhook-driven payment confirmation, TTL-based slot expiry, and Cloudinary-based media pipeline.
- Practiced Data Structures & Algorithms consistently on LeetCode to strengthen problem-solving and coding-interview readiness.
- Completing Docker certification to strengthen containerization and deployment skills.
- Freelancing on small client projects, independently delivering custom full-stack web solutions.

### PROJECTS

#### VoltSpot - EV Charging Station Booking Platform

[Live](#) | [GitHub](#)

React.js, Redux Toolkit, Tailwind CSS, Node.js, Express.js, MongoDB, Socket.IO, Stripe, Cloudinary, Nodemailer

- Architected a full-stack EV charging booking platform with three role-based modules (User, Partner, Admin), including an interactive **react-leaflet** map for station discovery and location pinning.
- Engineered a two-phase booking state machine — bookings held as "pending" via **MongoDB TTL indexes** and confirmed only through a verified **Stripe webhook** (checkout.session.completed) — eliminating double-booking and payment-state race conditions.
- Implemented **wallet and refund infrastructure**, including cancel/rebook flows, automatic wallet credit on cancellation, and a unified transaction history merging payments and refunds.
- Migrated image storage from local Multer disk storage to **Cloudinary** to support persistent uploads on Render's ephemeral filesystem, and resolved production email delivery by switching to **Brevo SMTP**.
- Delivered real-time features with **Socket.IO** for live booking/status updates, **JWT-based multi-role authentication**, and OTP-verified partner onboarding with auto-expiring unverified accounts.
- Built responsive, paginated admin dashboards (users, partners, stations) and a partner dashboard (profile, bookings, payments) using a consistent **Tailwind CSS** design system.

#### ChatDot - Real-Time Chat Application

[Live](#) | [GitHub](#)

React.js, Node.js, Express.js, Socket.IO, Tailwind CSS

- Built a real-time messaging application supporting instant delivery, dynamic UI updates, and live user-presence tracking.
- Implemented an event-driven **Socket.IO** architecture enabling bidirectional communication and automatic online-user-list synchronization.
- Designed a responsive **React and Tailwind CSS** frontend paired with an Express.js backend handling authentication and WebSocket connections.

#### Automated Plant Disease Detection System

Python, Django, SQL, Computer Vision

- Developed a machine-learning-based **computer vision** system to detect and classify plant diseases from images.
- Delivered real-time diagnosis with actionable treatment recommendations to reduce crop loss and improve agricultural productivity.

### EDUCATION

#### Bachelor of Computer Science — First Class

2021 – 2024

College of Applied Science, Pinarayi, Kannur, Kerala